

AI at the Frontier of Personalized Cancer Care

A summary of **ASCO innovations** at the intersection of artificial intelligence and oncology – where leading clinicians explore how AI is reshaping treatment decisions, clinical trial design, and the future of precision cancer medicine.



ASCO AI IN ONCOLOGY

Why AI, Why Now?

Artificial intelligence is no longer a distant promise – it is actively reshaping how clinicians identify therapeutic targets, match patients to tailored treatments, and design more efficient, smarter clinical trials. New tools and technologies are emerging at an unprecedented pace, each with the potential to accelerate the precision oncology agenda.

Target Identification

AI models analyze genomic and molecular data at scale to pinpoint actionable therapeutic targets with far greater speed and precision than traditional methods.

Patient–Therapy Matching

Machine learning algorithms help align the right patient to the right treatment, reducing trial-and-error and improving the probability of meaningful clinical response.

Smarter Trial Design

AI-enabled adaptive trial frameworks allow real-time protocol refinement, reducing patient burden and accelerating the path from hypothesis to evidence.

The Core Challenge: Promise vs. Practice

Despite rapid advances, a critical gap persists between AI's theoretical potential and its integration into everyday clinical workflows. Oncologists face real implementation, validation, and trust barriers that must be addressed before AI tools can meaningfully transform patient care at scale.

Implementation Barriers

Integrating AI into existing electronic health record systems and clinical workflows requires significant technical infrastructure and change management.

Validation Challenges

Many AI models lack rigorous prospective clinical validation, making it difficult for clinicians to assess reliability and generalizability across diverse patient populations.

Building Trust

Clinician trust in AI recommendations depends on transparency of model logic, explainability of outputs, and demonstrated alignment with established clinical standards.

Featured Speakers

This ASCO webinar convenes two recognized leaders at the forefront of AI-enabled cancer care, bridging clinical expertise, drug development, and health technology strategy.



Vivek Subbiah, MD
Jeffrey and Christina Bird
Professor of Medicine and
of Pediatrics

Stanford University School of
Medicine

Associate Director, Drug
Development & Precision
Oncology, Stanford Cancer
Institute; Executive Medical
Director, Novel Therapies &
Clinical Trial Network
Development, Stanford
Health Care



Chadi Nabhan, MD,
MBA, FACP
Chief Medical Officer and
Head of Strategy

Ryght AI, Inc.

Department of Clinical
Pharmacy and Outcomes
Sciences, University of South
Carolina

AI's Impact on Clinical Workflows

Beyond diagnosis and treatment selection, AI is redefining how oncology teams operate day-to-day – reducing administrative burden, surfacing relevant evidence, and enabling more informed and timely clinical decisions for every patient encounter.

Reducing Clinician Burden

AI automates time-consuming tasks – from documentation and prior authorization to literature review – freeing clinicians to focus on high-value patient interactions and complex decision-making.

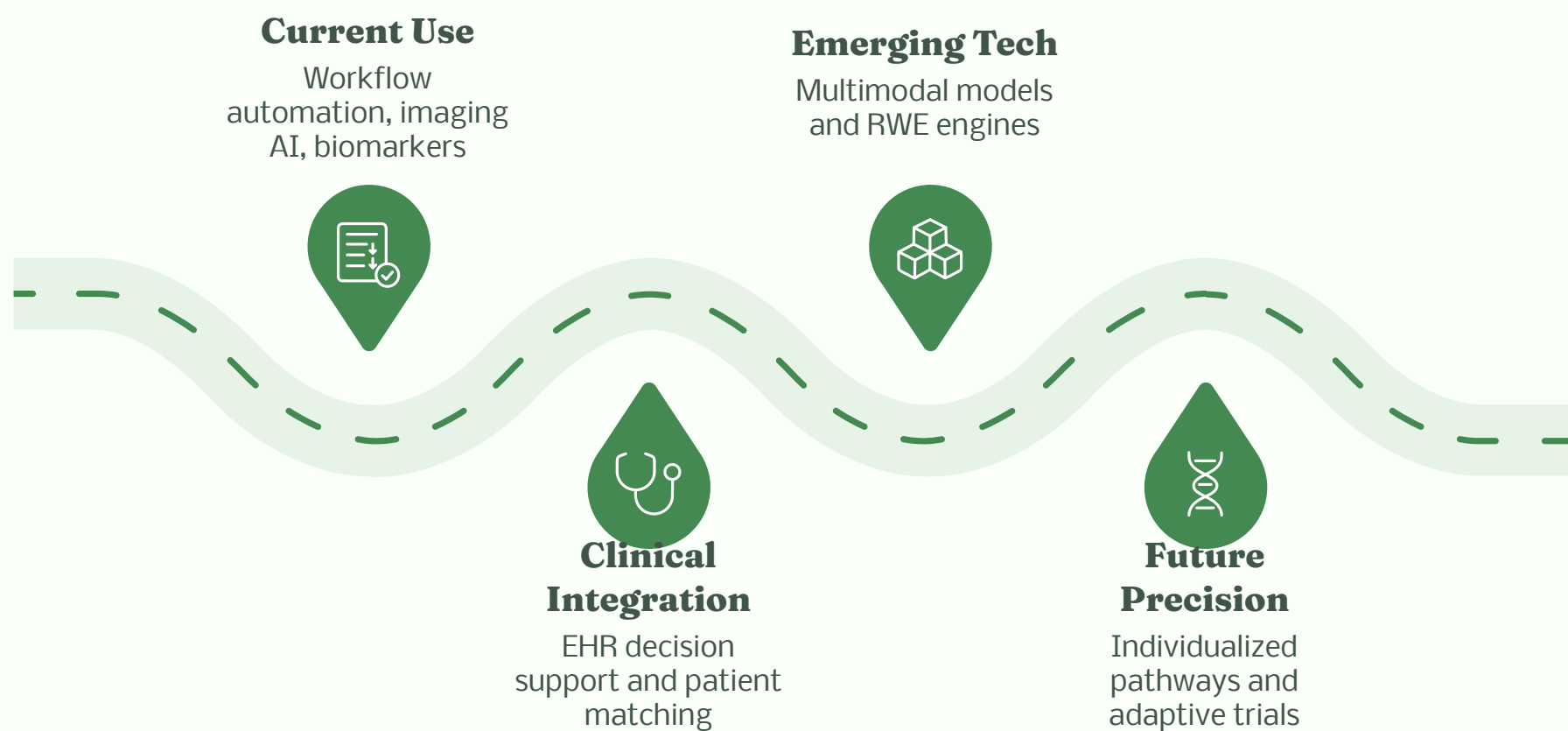
Deepening Patient & Disease Understanding

Advanced AI models integrate multi-omic, imaging, and clinical data to paint a richer picture of each patient's disease – revealing subtle patterns and predictive signals invisible to traditional analysis alone.

Enhancing Treatment Decisions

By synthesizing evidence from thousands of comparable cases, AI tools support more consistent, evidence-based therapy selection, reducing variation in care and improving outcomes at the population level.

From Current Use Cases to Emerging Technologies



The trajectory of AI in oncology moves from isolated point solutions toward deeply integrated, continuously learning systems capable of personalizing every dimension of cancer care – from screening and diagnosis through treatment selection and survivorship monitoring.

AI Literacy: A Non-Negotiable for Modern Oncologists

"AI will be a part of life going forward, and AI literacy is very important for physicians – especially in the counseling of patients."

– **Travis Zack, MD, PhD**, OpenEvidence & UCSF

Dr. Zack, speaking with **Jame Abraham, MD, FACP** of Cleveland Clinic for *ASCO AI in Oncology*, underscored that oncologists must become active participants – not passive observers – in this transformation.

Understanding how AI tools work, their limitations, and how to communicate AI-derived recommendations to patients is now a core clinical competency.

The Physician's Role in the AI Era

Clinicians must move beyond skepticism and passive observation to become informed stewards of AI tools. Active participation ensures that AI systems are deployed safely, equitably, and in ways that genuinely solve clinical problems.

01

Build AI Literacy

Understand the fundamentals of how AI models are trained, validated, and deployed – including their known limitations and potential failure modes in clinical contexts.

02

Engage in Validation

Participate in prospective studies and real-world evaluations that test AI tools against diverse patient populations, ensuring outputs are clinically meaningful and safe.

03

Counsel Patients Thoughtfully

Guide patients through AI-informed recommendations with transparency – explaining what the tool suggests, why, and how it fits into the broader clinical picture.

04

Shape the Future

Advocate for ethical, patient-centered AI development by contributing clinical expertise to tool design, regulatory feedback, and institutional AI governance frameworks.

Key Themes from ASCO AI in Oncology



Precision at Scale

AI enables truly individualized treatment strategies by integrating genomic, clinical, and imaging data across large patient cohorts – unlocking precision that was previously achievable only in research settings.



Safety & Trust First

Responsible AI deployment demands rigorous validation, transparent model design, and ongoing post-market surveillance to maintain patient safety and earn lasting clinician trust.



Multidisciplinary Collaboration

The most impactful AI innovations emerge from close collaboration between oncologists, data scientists, trialists, and patients – ensuring tools address real-world clinical needs.



Continuous Learning

AI literacy must be embedded in medical education and ongoing professional development so that tomorrow's oncologists are equipped to lead, evaluate, and responsibly deploy these technologies.

Moving from Promise to Personalized Oncology

The path forward requires bold clinical leadership, rigorous science, and a commitment to keeping patients at the center of every AI innovation. The tools are arriving – the question is whether the oncology community will rise to meet them with the expertise, curiosity, and responsibility they demand.

Embrace AI Literacy

Make understanding AI tools a professional priority – for yourself, your team, and your trainees.

Demand Rigorous Validation

Insist on prospective, diverse-population evidence before adopting AI tools in clinical decision-making.

Champion Patient Safety

Ensure every AI integration prioritizes transparency, equity, and demonstrable benefit for patients across all populations.



Follow **ASCO AI in Oncology** for ongoing expert conversations, emerging research, and practical guidance on integrating AI responsibly into your oncology practice.